Exam 1 Topics and Links

Lecture-4&5_Evaluation

- distance metrics
- Manhattan distance
- Euclidean distance
- Cosine similarity
- Jacard similarity
- Model Evaluation/validation
- Cross validation
- Overfitting
- Underfitting
- Performance metrics (Regression)
- Performance metrics (Classification)
- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1-Score
- Confusion matrix with multiple classes
- ROC Curve
- Area under the ROC Curve

Lecture-6_Review and Bayes Classifier

- Experiment
- Sample Space
- Event
- Probability addition and multiplication rules
- Conditional probability
- Independent Events
- Dependent Events
- Rule of total probability
- Bayes' Theorem:
- Random Variables
- PDF, PMF and CDF
- Joint Probability Distribution

- Probability Chain Rule
- Mean and Variance
- Covariance
- Information
- Entropy
- Joint Entropy
- Conditional Entropy
- Mutual Information
- Relative Entropy:
- Bayes Classifier
- Estimating the Prior Probability:
- Estimating the Likelihood:
- Naïve Bayes

Lecture-7_DT

- Rule-Based Classifier
- decision tree
- Gini impurity
- Feature Selection
- Information Gain
- Information Gain Ratio
- Prevent Overfitting in Decision Trees
- Pruning (Pre and post)

Lecture-8_Perceptron

- Perceptron
- Margin
- Linearly Separable
- Loss Function
- Gradient Descent
- Perceptron Algorithm with SGD
- Perceptron Convergence Theorem
- Perceptron Limitation

Lecture-9&10_Logistic&SVM

- Logistic Regression
- Sigmoid Function
- Gradient of Logistic Loss

- SVM: Hyperplane
- SVM: Optimization
- Lagrange and Dual Problem
- Soft-Margin SVM
- Slack Variables
- kernel function

Lecture-11_Boosting

- Ensemble Learning:
- Bagging:
- Random Forests
- Bias-Variance Tradeoff
- Boosting:
- AdaBoost

Lecture-12_Clustering

- Supervised Learning
- Unsupervised Learning
- Clustering
- Hard Clustering vs. Soft Clustering:
- K-means Clustering
- K-means
- picking K
- Mixtures of Gaussians/Gaussian Mixture Model
- Gaussian Distribution/Mixture Models

Lecture-14_t-SNE

- The Curse of Dimensionality
- Feature Selection
- Feature Extraction
- Dimensionality Reduction: PCA
- Nonlinear Dimensionality Reduction/t-SNE
- Data Visualization Gap
- Stochastic Neighbor Embedding (SNE)
- KL Divergence
- Hyperparameter: Perplexity

Lecture-4&5_Evaluation links

- Weird notions of "distance" || Intro to Metric Spaces
- ML 20 : Distance Metrics Models | Euclidean | Manhattan | Minkowski | Hamming Distance with Examples
- Cosine Similarity, Clearly Explained!!!
- Every Distance in Data Science (Almost 100K Subs!)
- Euclidean Distance and Manhattan Distance
- 5. Cosine Similarity vs. Euclidean Distance vs. Dot Product: Choosing the Right Metric for AI Search
- Cosine Distance vs Euclidean Distance in Machine Learning and NLP with Word2Vec or Glove Vectors
- Understand Cosine Similarity | 2 Minute Tutorial
- How to find Euclidean Manhattan Minkowski distance Supremum distance Cosine Similarity Mahesh Huddar
- Euclidean Distance & Cosine Similarity | Introduction to Data Mining | Part 17
- Euclidean Manhattan and Cosine Distance | Euclidean distance vs Cosine similarity
- Cosine similarity, cosine distance explained | Math, Statistics for data science, machine learning
- Distance and Similarity measures
- Machine Learning Fundamentals: Cross Validation
- K-Fold Cross Validation - Intro to Machine Learning
- How to evaluate ML models | Evaluation metrics for machine learning
- Why do we split data into train test and validation sets?
- Module 8. Introduction - Cross Validation; Hyperparameter tuning; Model Evaluation
- Cross Validation : Data Science Concepts
- Cross Validation
- Machine Learning and Cross-Validation
- Underfitting & Overfitting - Explained
- Machine Learning Fundamentals: Bias and Variance
- Overfitting, Underfitting, and Bad Data Are Ruining Your Predictive Models
- But What Is Overfitting in Machine Learning?
- Regularization in a Neural Network | Dealing with overfitting
- Overfitting and underfitting, explained intuitively

- Overfitting and Underfitting | Bias and Variance Tradeoff in Machine Learning | Clearly Explained!
- Solve your model's overfitting and underfitting problems - Pt.1 (Coding TensorFlow)
- Machine Learning-Bias And Variance In Depth Intuition| Overfitting Underfitting
- Bias-Variance Tradeoff : Data Science Basics
- Bias and Variance for Machine Learning | Deep Learning
- Never Forget Again! // Precision vs Recall with a Clear Example of Precision and Recall
- Performance Metrics, Accuracy,Precision,Recall And F-Beta Score Explained In Hindi|Machine Learning
- Precision, Recall, & F1 Score Intuitively Explained
- How to Evaluate Your ML Models Effectively? | Evaluation Metrics in Machine Learning!
- Evaluation Metrics For Regression - When & Why To Use What
- Introduction to Regression analysis
- Regression Metrics in Machine Learning | MAE, MSE, RMSE, R2 Score(R-Squared) 📊🔍
- Precision, Recall, F1 score, True Positive|Deep Learning Tutorial 19 (Tensorflow2.0, Keras & Python)
- TP, FP, TN, FN, Accuracy, Precision, Recall, F1-Score, Sensitivity, Specificity, ROC, AUC
- Introduction to Precision, Recall and F1 - Classification Models | | Data Science in Minutes
- Machine Learning Fundamentals: The Confusion Matrix
- The Confusion Matrix in Machine Learning
- Confusion Matrix | How to evaluate classification model | Machine Learning Basics
- Confusion Matrices - Intro to Machine Learning
- What is a Confusion Matrix?
- Confusion Matrix | ML | True Positive | True Negative | False Positive | False Negative - P1
- Confusion to Clarity: Mastering Confusion Matrix in Machine Learning
- Evaluating Machine Learning Models
- The Confusion Matrix : Data Science Basics
- 145 - Confusion matrix, ROC and AUC in machine learning

- Confusion Matrix Explained
- Introduction to AI, Lesson 6.1 - Multiclass Confusion Matrix
- Confusion Matrix Solved Example Accuracy Precision Recall F1 Score Prevalence by Mahesh Huddar
- ROC and AUC, Clearly Explained!
- ROC Curve and AUC Value
- The ROC Curve : Data Science Concepts
- ROC Curves and Area Under the Curve (AUC) Explained
- A Critical Skill People Learn Too LATE: Learning Curves In Machine Learning.
- ROC curve and AUC metric in 5 minutes
- Receiver Operating Characteristic (ROC) Curves | Applied Machine Learning, Part 2
- ROC (Receiver Operating Characteristic) Curve in 10 minutes!
- ROC Curve - how to select the BEST threshold
- 145 - Confusion matrix, ROC and AUC in machine learning
- Tutorial 41-Performance Metrics(ROC,AUC Curve) For Classification Problem In Machine Learning Part 2
- Introduction to AI, Lesson 6.1 - Multiclass Confusion Matrix

Lecture-6_Review and Bayes Classifier

- The Law of Total Probability | Probability Theory, Total Probability Rule
- The Law of Total Probability
- 4.2.3 Law of Total Probability: Video
- Bayes theorem, the geometry of changing beliefs
- The Law of Total Probability
- Bayes' Theorem - The Simplest Case
- Intro to Bayes's Theorem | Probability Theory
- Machine Learning: Bayes Decision Theory
- Bayes Theorem in 3 Minutes
- Naive Bayes, Clearly Explained!!!
- Gaussian Naive Bayes, Clearly Explained!!!
- Probability Distributions Clearly Explained Visually (PMF, PDF and CDF)
- Probability density and mass functions
- Probability Distribution Functions (PMF, PDF, CDF)
- Basics of joint probability

- Marginal, Joint and Conditional Probabilities Explained
- Basic probability: Joint, marginal and conditional probability | Independence
- The Chain Rule for Probability
- Chain Rule in Probability
- The Chain Rule, Clearly Explained!!!
- Sum Rule, Product Rule, Joint & Marginal Probability - CLEARLY EXPLAINED with EXAMPLES!
- Covariance Clearly Explained!
- Machine Learning Fundamentals: Bias and Variance
- The Covariance Matrix : Data Science Basics
- Covariance Matrix - Explained
- Entropy (for data science) Clearly Explained!!!
- Information Theory Basics
- Intuitively Understanding the Shannon Entropy
- Information entropy | Journey into information theory | Computer Science | Khan Academy
- Information Theory, Lecture 1: Defining Entropy and Information - Oxford Mathematics 3rd Yr Lecture
- A Short Introduction to Entropy, Cross-Entropy and KL-Divergence
- Introduction to Information Theory: Entropy - Part 6 - Conditional Entropy
- Mutual Information, Clearly Explained!!!
- The KL Divergence : Data Science Basics
- Entropy (for data science) Clearly Explained!!!
- Intuitively Understanding the KL Divergence
- The Key Equation Behind Probability
- Intuitively Understanding the Cross Entropy Loss
- Naive Bayes, Clearly Explained!!!
- Tutorial 48- Naive Bayes' Classifier Indepth Intuition- Machine Learning
- The Math Behind Bayesian Classifiers Clearly Explained!
- Machine Learning: Bayes Decision Theory
- Naive Bayes classifier: A friendly approach
- 11. Naïve Bayesian Classification Numerical Solved Example in Machine Learning by Mahesh Huddar
- Naive Bayes Classifier | Part 1 | Conditional Probability
- IAML5.1: Overview

- Lecture 5 - GDA & Naive Bayes | Stanford CS229: Machine Learning Andrew Ng (Autumn 2018)

Lecture-7_DT

- Machine Learning | Rule-based Classifier
- #20 Rule Based Classifier with Example |DM|
- Rule-Based Learning as an Alternative to Machine Learning
- Decision and Classification Trees, Clearly Explained!!!
- 3. Reasoning: Goal Trees and Rule-Based Expert Systems
- Rule-Based Classifiers
- Decision Tree Classification Clearly Explained!
- Decision Tree: Important things to know
- Let's Write a Decision Tree Classifier from Scratch - Machine Learning Recipes #8
- Visual Guide to Decision Trees
- MIT: Machine Learning 6.036, Lecture 12: Decision trees and random forests (Fall 2020)
- Lecture 9 - Decision Trees and Ensemble Methods | Stanford CS229: Machine Learning (Autumn 2018)
- Decision Trees - VisuallyExplained
- Machine Learning Lecture 29 "Decision Trees / Regression Trees" -Cornell CS4780 SP17
- Regression Trees, Clearly Explained!!!
- Machine Intelligence - Lecture 16 (Decision Trees)
- All Major Feature Selection Methods in Machine Learning Explained
- https://www.youtube.com/watch?v=YNgP05M_HVc
- StatQuest: Decision Trees, Part 2 - Feature Selection and Missing Data
- Entropy (for data science) Clearly Explained!!!
- Information Gain Calculation Part 1 - Intro to Machine Learning
- Information Gain
- How to find the Entropy and Information Gain in Decision Tree Learning by Mahesh Huddar
- Tutorial 38- Decision Tree Information Gain
- Underfitting & Overfitting - Explained
- How to Avoid Overfitting in Decision Tree Learning | Machine Learning | Data Mining by Mahesh Huddar
- Prune a decision tree to avoid overfitting

- IAML7.7 Overfitting in decision trees
- Decision Trees
- How to Prune Regression Trees, Clearly Explained!!!
- Decision Tree Pruning
- Decision Tree Pruning explained (Pre-Pruning and Post-Pruning)
- Pruning in Decision Trees
- Gini Index and Entropy|Gini Index and Information gain in Decision Tree|Decision tree splitting rule

Lecture-8_Perceptron

- Perceptron | Neural Networks
- Perceptron
- The Perceptron Explained
- ChatGPT is made from 100 million of these [The Perceptron]
- Perceptron Algorithm with Code Example - ML for beginners!
- What are MLPs (Multilayer Perceptrons)?
- Perceptrons: The First Trainable Neural Networks | Teaching Computers to Learn, Part 3
- Linearly Separable Data - Intro to Machine Learning
- Soft Margin SVM : Data Science Concepts
- Support Vector Machine (SVM) in 2 minutes
- IR20.2 Large margin classification
- 20. Linearly separable- Neural networks
- #Linear separability in neural networks|Linearly separable
- Support Vector Machines: All you need to know!
- ECE595ML Lecture 06-1 Linear Separability
- Gradient Descent, Step-by-Step
- Gradient descent, how neural networks learn | Deep Learning Chapter 2
- Gradient Descent in 3 minutes
- Intro to Gradient Descent || Optimizing High-Dimensional Equations
- Gradient Descent Explained
- STOCHASTIC Gradient Descent (in 3 minutes)
- Who's Adam and What's He Optimizing? | Deep Dive into Optimizers for Machine Learning!
- Optimization for Deep Learning (Momentum, RMSprop, AdaGrad, Adam)
- Perceptron Training

- Perceptron Explained | What It Is & How the Perceptron Convergence Algorithm Works
- Lecture 6 "Perceptron Convergence Proof" -Cornell CS4780 SP17
- Perceptrons V Neurons - Machine Learning Is Not Like Your Brain #2
- [Neural Network 3] Limitations of Perceptron and Multilayer Perceptron
- Lecture 5 "Perceptron" -Cornell CS4780 SP17

Lecture-9&10_Logistic&SVM

- Logistic Regression in 3 Minutes
- StatQuest: Logistic Regression
- Logistic Regression Details Pt1: Coefficients
- Logistic Regression Details Pt 2: Maximum Likelihood
- Logistic Regression Details Pt 3: R-squared and p-value
- Logistic Regression (and why it's different from Linear Regression)
- Machine Learning Crash Course: Logistic Regression
- Machine Learning and Logistic Regression
- Linear Regression vs Logistic Regression - What's The Difference?
- MIT: Machine Learning 6.036, Lecture 4: Logistic regression (Fall 2020)
- Locally Weighted & Logistic Regression | Stanford CS229: Machine Learning - Lecture 3 (Autumn 2018)
- Logistic Regression Part 1 | Perceptron Trick
- Explaining logistic regression | VNT #14
- Machine Learning Tutorial Python - 8: Logistic Regression (Binary Classification)
- Support Vector Machine (SVM) in 2 minutes
- Support Vector Machines: All you need to know!
- Support Vector Machines : Data Science Concepts
- SVM (The Math) : Data Science Concepts
- SVM Kernels : Data Science Concepts
- SVM Dual : Data Science Concepts
- Support Vector Machines Part 1 (of 3): Main Ideas!!!
- 16. Learning: Support Vector Machines
- Support Vector Machines - THE MATH YOU SHOULD KNOW
- SVM Kernels : Data Science Concepts

- The Kernel Trick
- Linear Algebra for Machine Learning: Line, Plane, Hyper Plane :@ Applied AI Course
- SVM Dual : Data Science Concepts
- Convexity and The Principle of Duality
- Duality: Lagrangian and dual problem
- Lagrange Multipliers | Geometric Meaning & Full Example
- Constrained Optimization: Intuition behind the Lagrangian
- Soft Margin SVM : Data Science Concepts
- Soft-Margin SVM
- Soft SVM - Soft Support Vector Machine - Machine Learning
- The Kernel Trick - THE MATH YOU SHOULD KNOW!
- The Kernel Trick
- Support Vector Machines Part 2: The Polynomial Kernel (Part 2 of 3)
- Support Vector Machines: Slack Variables
- Support Vector Machines (SVM) - the basics | simply explained
- SVM (The Math) : Data Science Concepts
- Support Vector Machines - THE MATH YOU SHOULD KNOW
- Support Vector Machines (SVM) Hard Margin Dual Formulation - Math Explained Step By Step
- SVM algorithm Find Hyperplane Solved Numerical Example in Machine Learning by Mahesh Huddar
- 16. Learning: Support Vector Machines

Lecture-11_Boosting

- Bagging vs Boosting - Ensemble Learning In Machine Learning Explained
- Ensemble learners
- Machine Learning Tutorial Python - 21: Ensemble Learning - Bagging
- Ensemble (Boosting, Bagging, and Stacking) in Machine Learning: Easy Explanation for Data Scientists
- Master Ensemble Models: Bagging vs Boosting in Machine Learning EXPLAINED
- Intuition: Ensemble Learning | Ensembles in Machine Learning #1
- Lecture 9 - Decision Trees and Ensemble Methods | Stanford CS229: Machine Learning (Autumn 2018)

- AdaBoost, Clearly Explained
- StatQuest: Random Forests Part 1 - Building, Using and Evaluating
- 17. Learning: Boosting
- Gradient Boosting : Data Science's Silver Bullet
- Gradient Boost Part 1 (of 4): Regression Main Ideas
- Random Forests : Data Science Concepts
- StatQuest: Random Forests Part 1 - Building, Using and Evaluating
- StatQuest: Random Forests Part 2: Missing data and clustering
- Introduction to Ensemble learning|What is Ensemble Modelling|Understanding Ensemble Models
- Bagging Vs Boosting In Machine Learning | Ensemble Learning In Machine Learning | Simplilearn
- Visual Guide to Gradient Boosted Trees (xgboost)
- Machine Learning Fundamentals: Bias and Variance
- AdaBoost, Clearly Explained
- What is AdaBoost (BOOSTING TECHNIQUES)
- Boosting
- AdaBoost for Beginners | Machine Learning in Plain English
- Extending Machine Learning Algorithms – AdaBoost Classifier | packtpub.com
- AdaBoost Math with Example clearly explained Step by Step Machine Learning Ensembles
- AdaBoost : Data Science Concepts
- Machine Learning Lecture 32 "Boosting" -Cornell CS4780 SP17

Lecture-12_Clustering

- Supervised vs. Unsupervised Learning
- Supervised vs Unsupervised vs Reinforcement Learning | Machine Learning Tutorial | Simplilearn
- Supervised & Unsupervised Machine Learning
- Supervised Learning: Crash Course AI #2
- Unsupervised Learning: Crash Course AI #6
- Supervised vs. Unsupervised Machine Learning: What's the Difference?
- Machine Learning Basics: Supervised v Unsupervised
- StatQuest: K-means clustering
- 12. Clustering
- Clustering in Machine Learning

- StatQuest: Hierarchical Clustering
- 4 Basic Types of Cluster Analysis used in Data Analytics
- Clustering: K-means and Hierarchical
- What is Clustering | Introduction to Machine Learning for beginners | The Most Used Terms 10
- Machine Learning Tutorial Python - 13: K Means Clustering Algorithm
- Complete Linkage Hierarchical Clustering Agglomerative Clustering in ML by Vidya Mahesh Huddar
- What are Gaussian Mixture Models? | Soft clustering | Unsupervised Machine Learning | Data Science
- Soft Clustering - Georgia Tech - Machine Learning
- Clustering 2: soft vs. hard clustering
- Fuzzy C Means Clustering Algorithm Solved Example | Clustering Algorithm in ML & DL by Mahesh Huddar
- Tutorial 33: Hard clustering VS Soft Clustering | What is Clustering and its different algorithms
- K-Means Clustering Explanation and Visualization
- What is KMeans Clustering? - A Quick Introduction to the Machine Learning Method
- How K-Means Clustering REALLY Works (Visual Explanation)
- K-Means Clustering Algorithm with Python Tutorial
- k-Means Cluster Analysis
- What are Gaussian Mixture Models? | Soft clustering | Unsupervised Machine Learning | Data Science
- Gaussian Mixture Models (GMM) Explained
- Gaussian Mixture Model
- Gaussian Mixture Model | Gaussian Mixture Model in Machine Learning | GMM Explained | Simplilearn
- Clustering (4): Gaussian Mixture Models and EM
- (ML 16.6) Gaussian mixture model (Mixture of Gaussians)
- StatQuest: K-nearest neighbors, Clearly Explained
- EM Algorithm : Data Science Concepts
- Gaussian Mixture Models

Lecture-14_t-SNE

- The Curse of Dimensionality
- Curse of Dimensionality - Georgia Tech - Machine Learning

- Curse of Dimensionality : Data Science Basics
- Machine Learning - Dimensionality Reduction - Feature Extraction & Selection
- Feature Selection in Machine Learning
- Feature Selection Techniques Easily Explained | Machine Learning
- Python Tutorial: Feature selection vs feature extraction
- Feature Selection in Machine Learning: Easy Explanation for Data Science Interviews
- Feature Selection In Machine Learning | Feature Selection Techniques With Examples | Simplilearn
- Feature Selection | Wrapper | Filter | Embeded Intrinsic Method in Machine Learning by Mahesh Huddar
- Class 27 Video: Feature Extraction and Machine Learning
- Principal Component Analysis (PCA) Explained: Simplify Complex Data for Machine Learning
- StatQuest: PCA main ideas in only 5 minutes!!!
- Principal Component Analysis (PCA)
- Dimensionality Reduction : Data Science Concepts
- StatQuest: Principal Component Analysis (PCA), Step-by-Step
- PCA Indepth Geometric And Mathematical InDepth Intuition ML Algorithms
- Principal Component Analysis (PCA)
- Principal Component Analysis (PCA)
- 17: Principal Components Analysis_ - Intro to Neural Computation
- Lecture 4 "Curse of Dimensionality / Perceptron" -Cornell CS4780 SP17
- What is t-SNE(T-distributed Stochastic Neighborhood Embedding) Lecture 21@Applied AI Course
- StatQuest: t-SNE, Clearly Explained
- t-SNE Simply Explained
- Latent Space Visualisation: PCA, t-SNE, UMAP | Deep Learning Animated
- Why T-distributed Stochastic Neighbor Embedding (TSNE) is great for visualization? Math Step By Step
- t-distributed Stochastic Neighbor Embedding (t-SNE) | Dimensionality Reduction Techniques (4/5)
- Intuitively Understanding the KL Divergence
- The KL Divergence : Data Science Basics

- Fantastic KL Divergence and How to (Actually) Compute It
- KL Divergence - How to tell how different two distributions are
- Language Model Evaluation and Perplexity
- Parameters vs hyperparameters in machine learning
- What is perplexity?
- What is Perplexity for LLMs?
- Train/Dev/Test Sets (C2W1L01)
- Hyperparameter Tuning Using Machine Learning Pipelines